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Course Code: DSA 0601
Course: Data Handling And Visualization

SET 26: STUDENT PERFORMANCE ANALYSIS

Q1. Histogram and Boxplot

Aim: visualize Math score distribution and compare Science scores by gender.

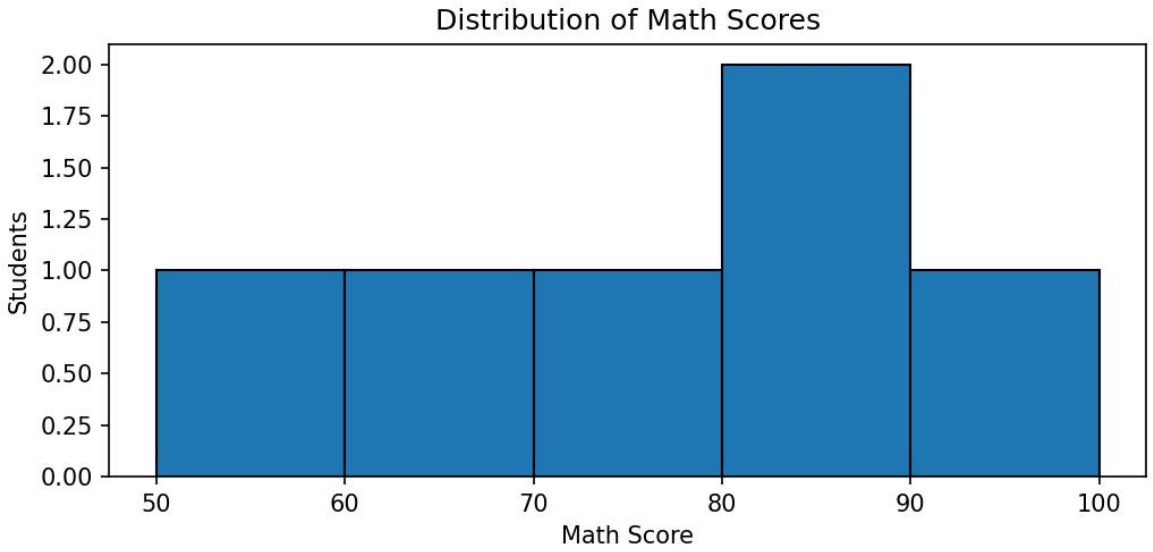
Algorithm:

1. Import Student_Mini_Data.csv.
2. Draw a histogram of Math_Score.
3. Draw Science_Score boxplots by Gender.
4. Add titles and labels.

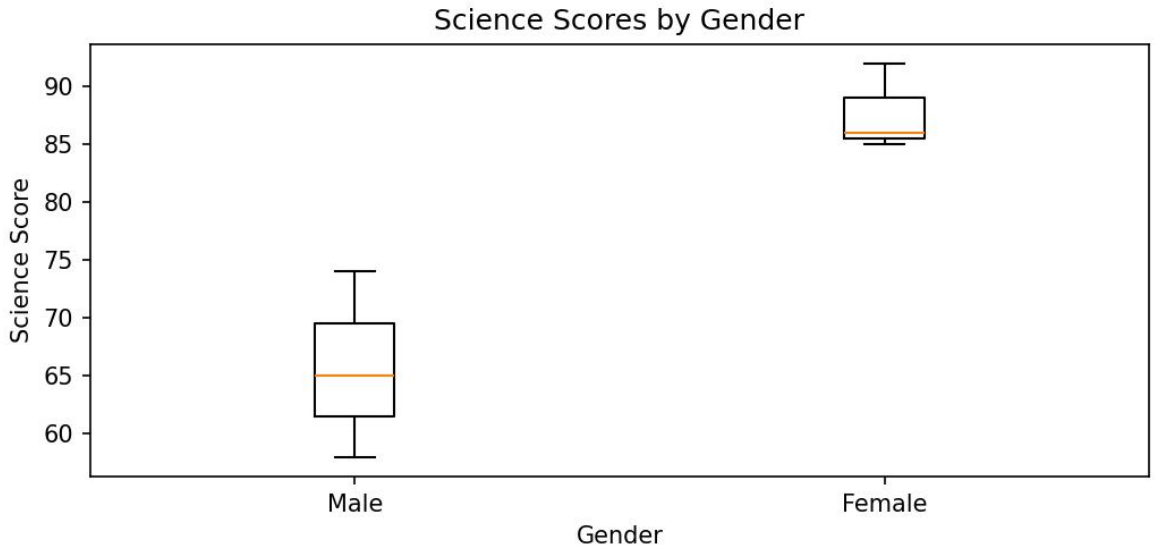
R Code:

```
students <- read.csv("Student_Mini_Data.csv")
students$Exam_Date <- as.Date(students$Exam_Date)
hist(students$Math_Score, col="skyblue", main="Distribution of Math Scores", xlab="Math Score",
ylab="Students")
boxplot(Science_Score ~ Gender, data=students, col=c("lightgreen","orange"), main="Science Scores by
Gender", xlab="Gender", ylab="Science Score")
```

Math Histogram:



Science Boxplot:



Result: Math scores range from 55 to 90. The female students' average Science score is higher in this six-student sample.

Q2. Study Hours vs Math Score

Aim: examine the relationship between study time and Math performance.

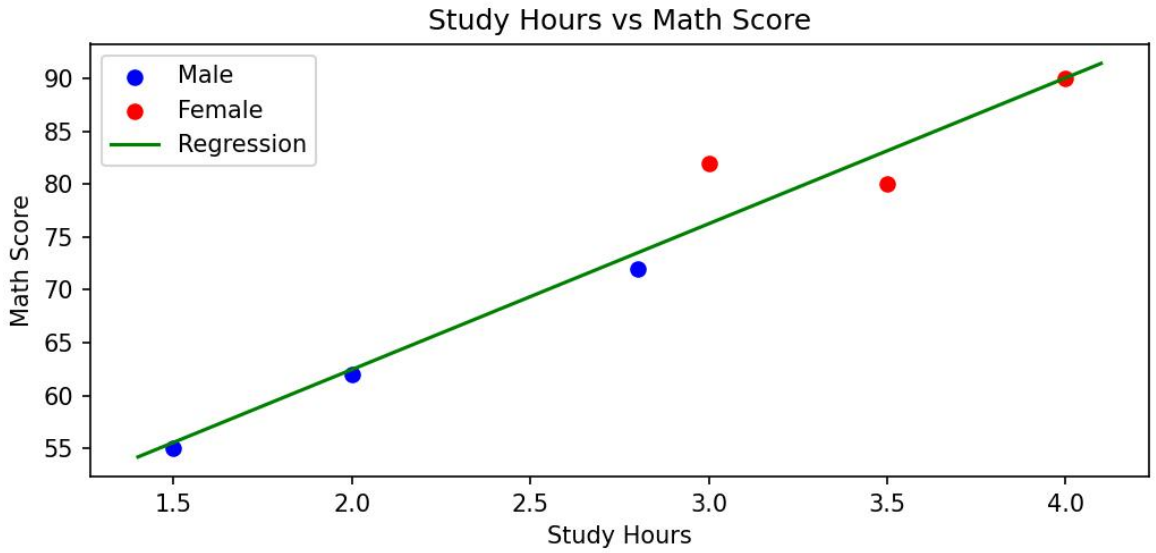
Algorithm:

1. Plot Study_Hours against Math_Score.
2. Color points by Gender.
3. Add a regression line and legend.

R Code:

```
plot(students$Study_Hours, students$Math_Score, col=ifelse(students$Gender=="Female","red","blue"),
pch=19, main="Study Hours vs Math Score", xlab="Study Hours", ylab="Math Score")
abline(lm(Math_Score ~ Study_Hours, data=students), col="darkgreen", lwd=2)
legend("topleft",c("Female","Male"),col=c("red","blue"),pch=19,bty="n")
```

Output Visualization:



Result: The sample shows a positive relationship between study hours and Math scores, but it does not prove causation.

Q3. Monthly Average and Moving Average

Aim: calculate monthly average Math scores and smooth the trend.

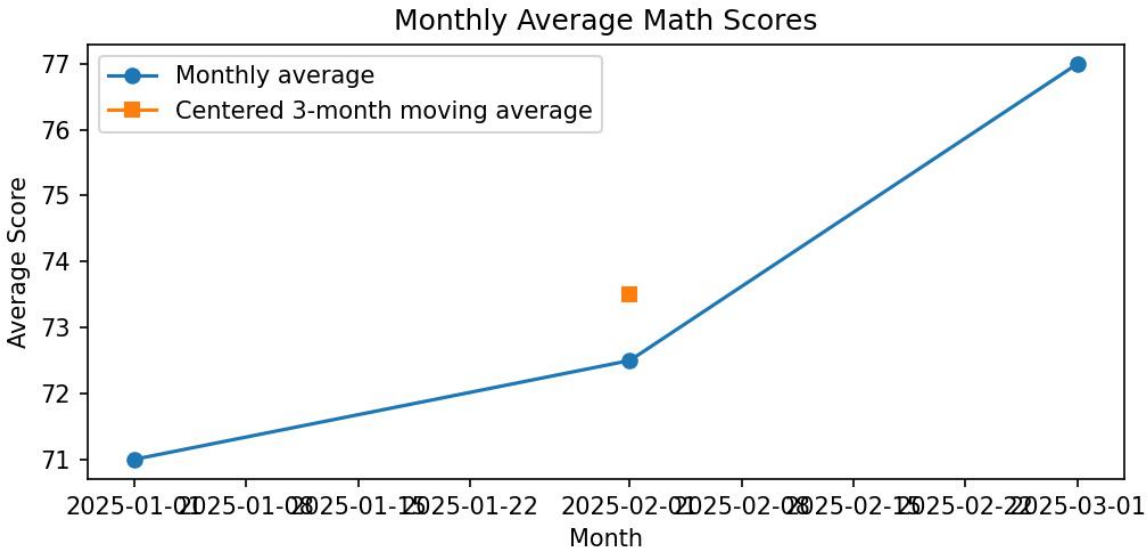
Algorithm:

1. Convert Exam_Date to Date.
2. Group records by month and calculate mean Math_Score.
3. Calculate centered three-month moving average.
4. Plot both series.

R Code:

```
students$Exam_Date <- as.Date(students$Exam_Date)
students$Month <- format(students$Exam_Date,"%Y-%m")
monthly <- aggregate(Math_Score ~ Month,data=students,mean)
monthly$Month_Date <- as.Date(paste0(monthly$Month,"-01"))
monthly$Moving_Average <- as.numeric(stats::filter(monthly$Math_Score,rep(1/3,3),sides=2))
print(monthly)
plot(monthly$Month_Date,monthly$Math_Score,type="b",pch=19,col="blue",main="Monthly Average Math
Scores",xlab="Month",ylab="Average Score")
lines(monthly$Month_Date,monthly$Moving_Average,type="b",pch=17,col="red")
```

Output Visualization:



Result: Monthly averages are January 71.0, February 72.5 and March 77.0, showing an increasing pattern. Only one centered three-month moving-average value is available.

Q4. Tableau Dashboard

Aim: compare average Math scores by gender and examine score trends over time.

Algorithm:

1. Connect Tableau to Student_Mini_Data.csv.
2. Create a bar chart of AVG(Math_Score) by Gender.
3. Create a line chart by Exam_Date.
4. Add an Exam_Date filter and combine sheets into a dashboard.

Result: The dashboard supports gender comparison and date-based exploration of student scores.